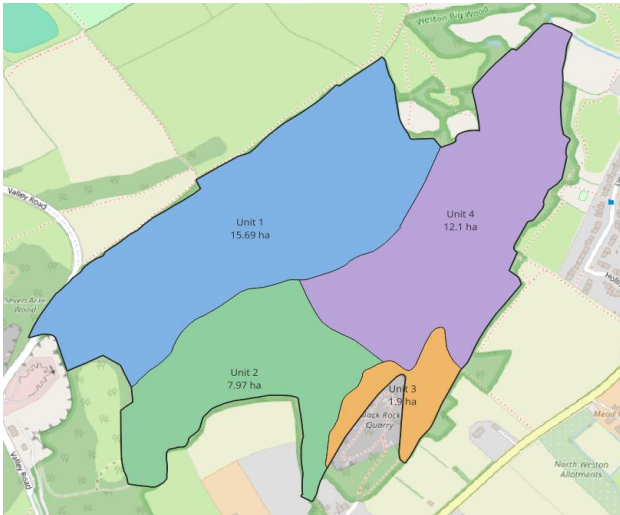


Weston Big Wood SSSI

Kaspar Szpojnarowicz

Few notes

1.



Natural England split the SSSI into four management units and assess each separately.

All four are the same habitat type, all four were assessed unfavorable and Declining in April 2023, and all four fail for the ash dieback and a lack of standing deadwood.

They are alike enough, and each has too few transects to stand on its own, so I combined them and look at the whole wood

- Ive done a lot of work in qgis, however most of the figures here are made in python so if/when I get more data I can keep working on it. Its quite messy but I can send you over a geopackage but not sure how itll do with the conversion from QGIS to ARCGIS
- Ive also made heat maps for lots of the map figures like you suggested but they aren't great due to not enough survey points

1. Site map

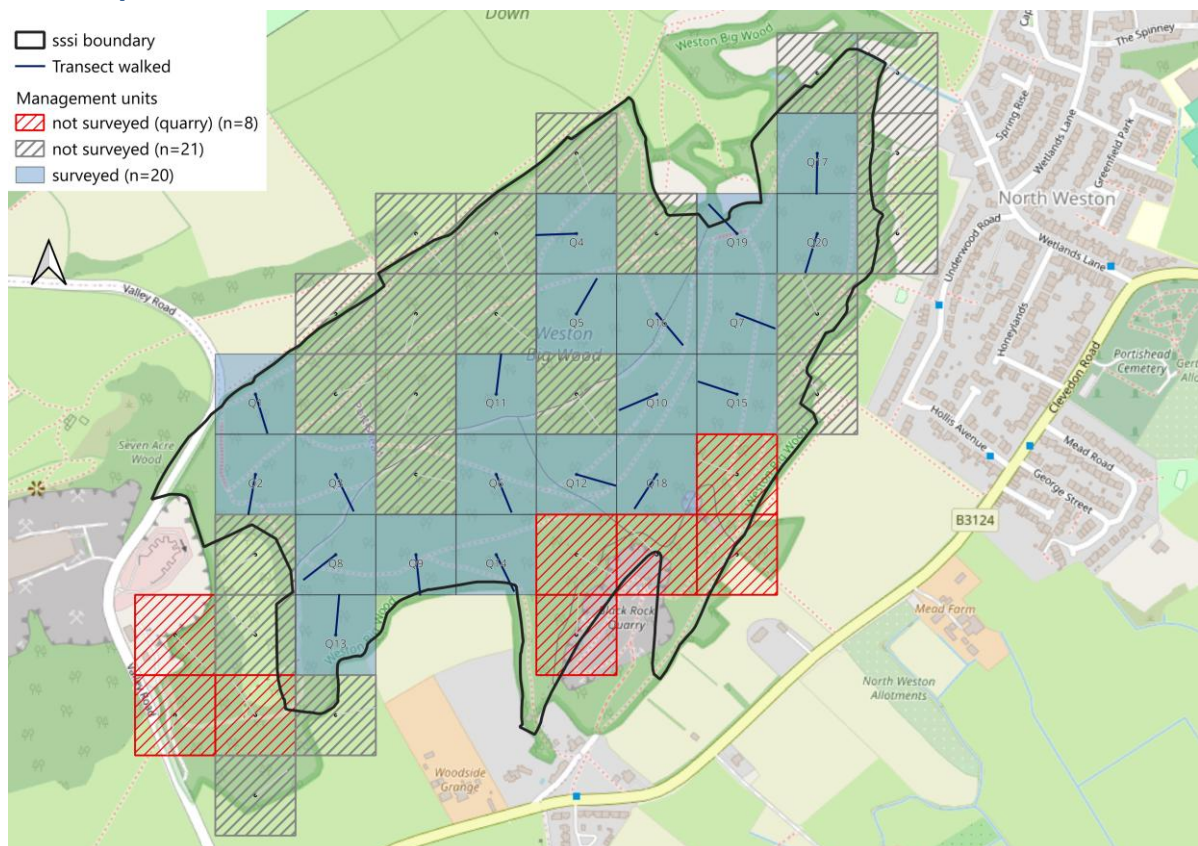


Figure 1 was made in QGIS. Red squares contain ground steeper than 60°, mostly quarry faces, checked against Environment Agency 1 m LIDAR and backed up with google earth. Only 8 of the 29 unsurveyed squares do, and 1 of the 20 surveyed. It would be interesting to go back and get the rest of the data

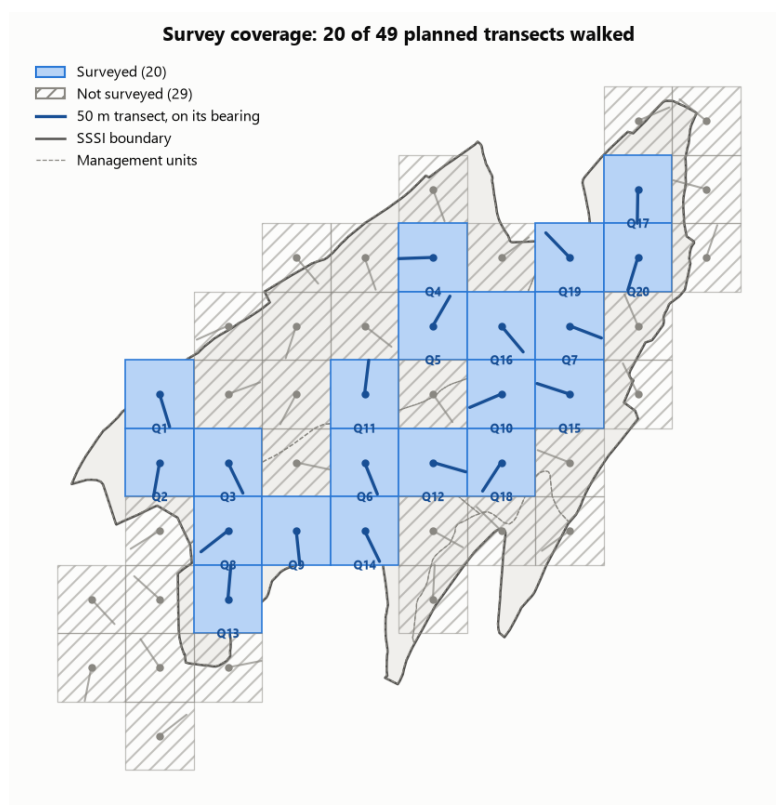


Figure 2. The same coverage without a basemap and made in python with each transect drawn on its planned bearing.

Notes

20 of the 49 were surveyed. That is 0.40 hectares, or 1.06% of the site. No transect falls in management unit 3.

Also, the reserve boundary is quite different to the SSSI boundary, especially in the bottom left corner.

2. Tree density

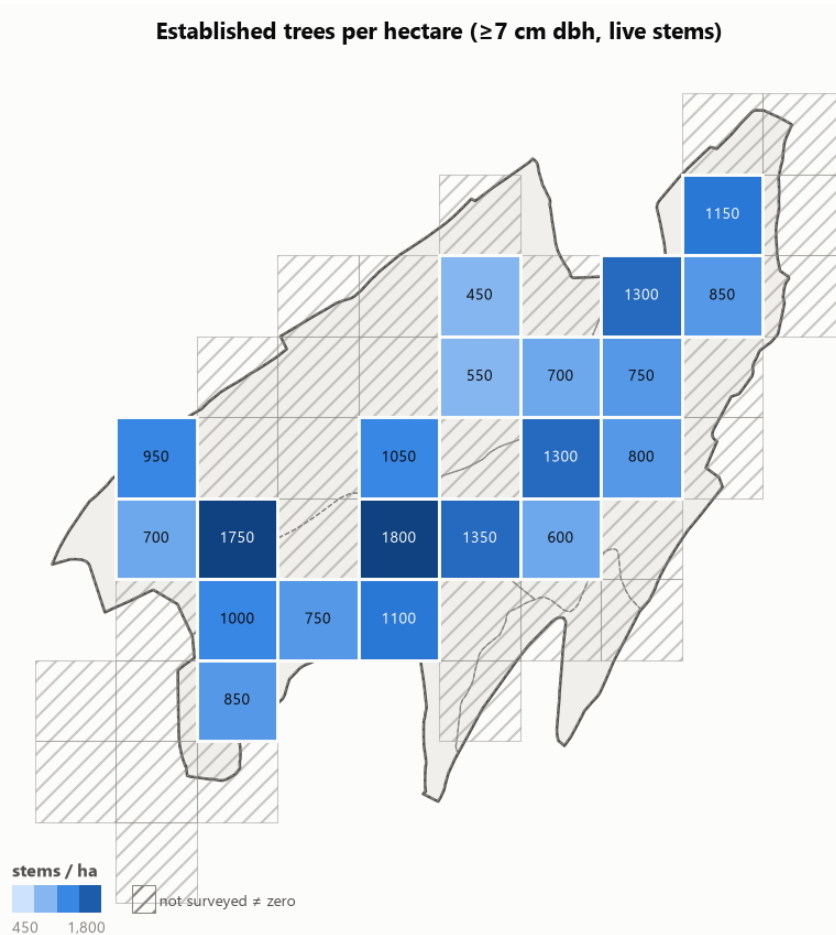


Figure 3 shows established trees per hectare, meaning stems of 7 cm diameter or more. Each square is one 200 m² transect scaled up to a hectare, which is a long stretch from a small sample.

The site averages 988 stems per hectare, but individual transects range from 450 to 1,800, so density varies fourfold across the wood.

Seedlings are left out. Field counts stopped at 100 per cell, so including them would give a floor rather than a real number, and they would swamp everything else.

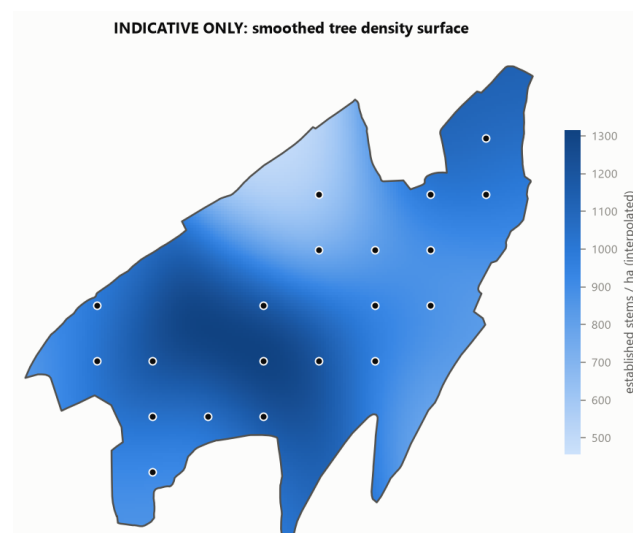


Figure 4. The same data smoothed. Indicative only: values between the sample points (white dots) are interpolated, not measured.

3. Species diversity

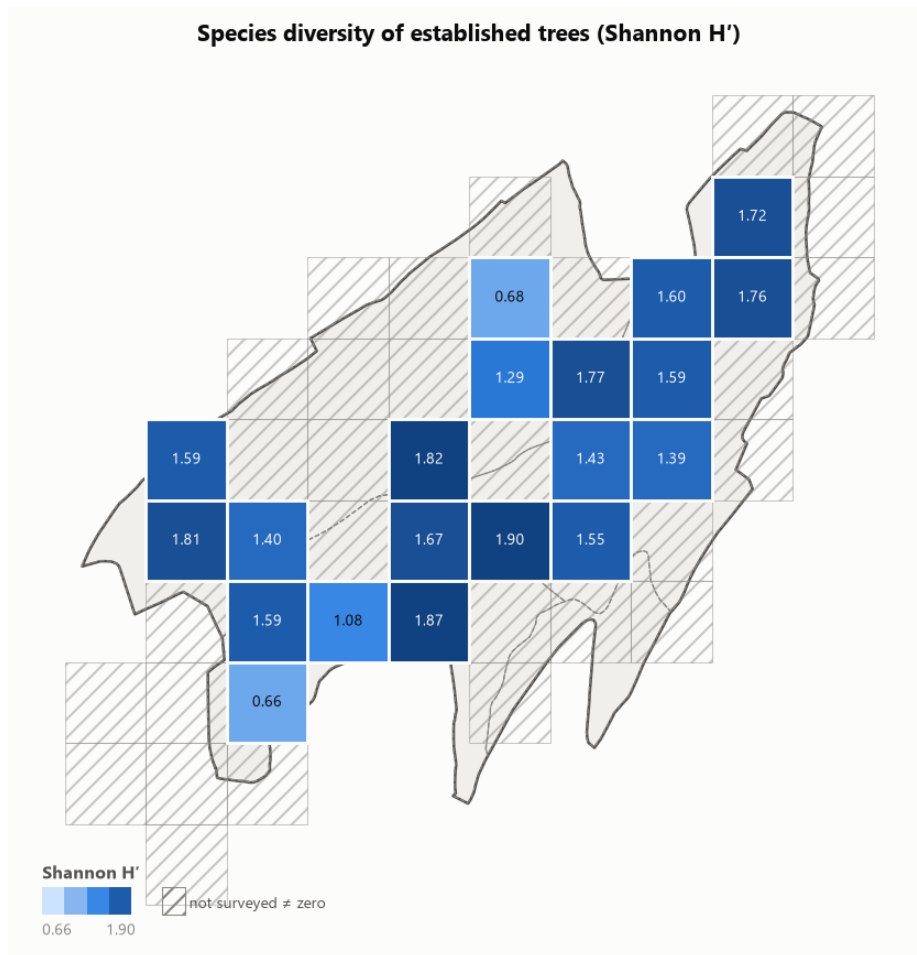


Figure 5 shows species diversity of established trees, measured as Shannon diversity. It rewards both the number of species present and how evenly the stems are spread between them, so a square with five species in roughly equal numbers scores higher than one with five species where nine stems in ten are hazel.

Values run from 0.66 to 1.90, and species richness from 3 to 9 species per transect.

The least diverse squares are not the emptiest ones. Q13 holds 17 established stems but 14 of them are hazel, which is what long-abandoned coppice looks like.

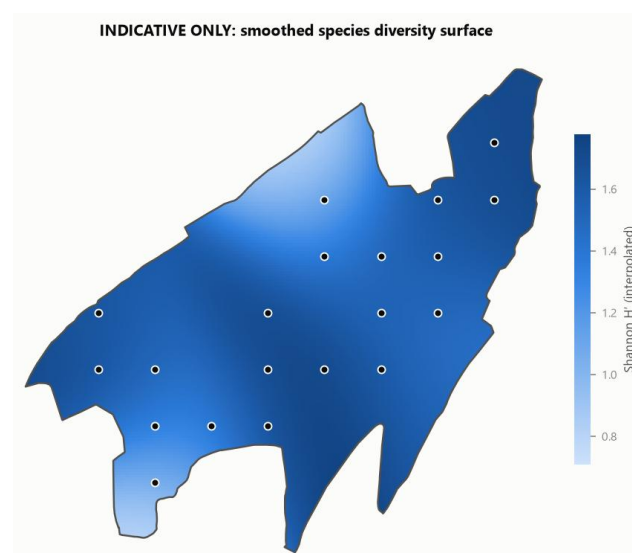


Figure 6. The same data smoothed. Indicative only.

4. Structural diversity

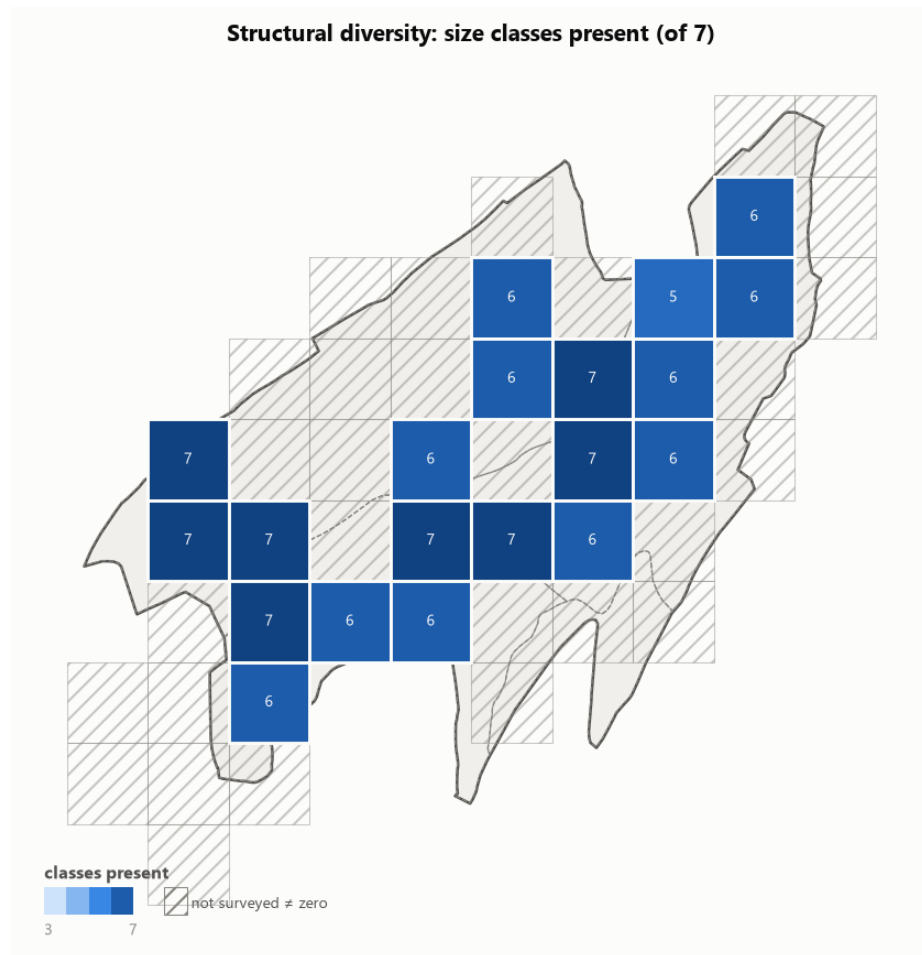


Figure 7 counts how many of the seven size classes are present in each transect, as a rough measure of how mixed-age the wood is.

Every surveyed transect holds 5 to 7 of the 7 classes, which looks healthy. Natural England's 2023 assessment reached the same conclusion, noting that "all age classes of trees are present".

I don't think that test tells us much, though. A single seedling counts the same as four hundred, and the seedling bank fills the smallest classes almost everywhere. Split the same data by species and the classes are missing in the trees that matter, **but I'll go into this later in my analysis.**

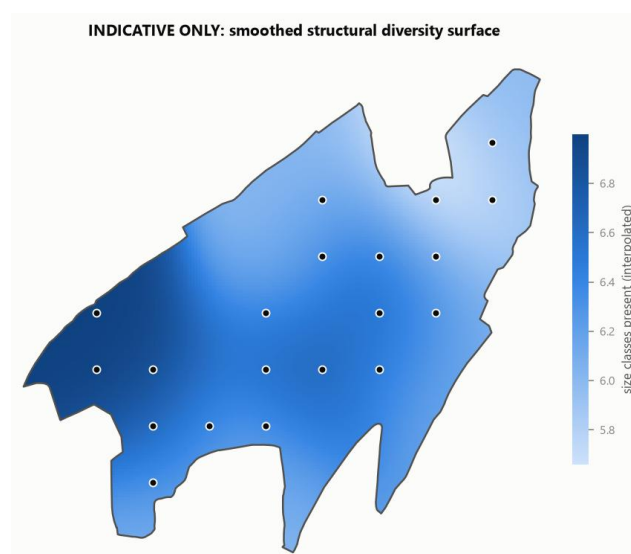


Figure 8. The same data smoothed. Indicative only.

5. Deadwood distribution

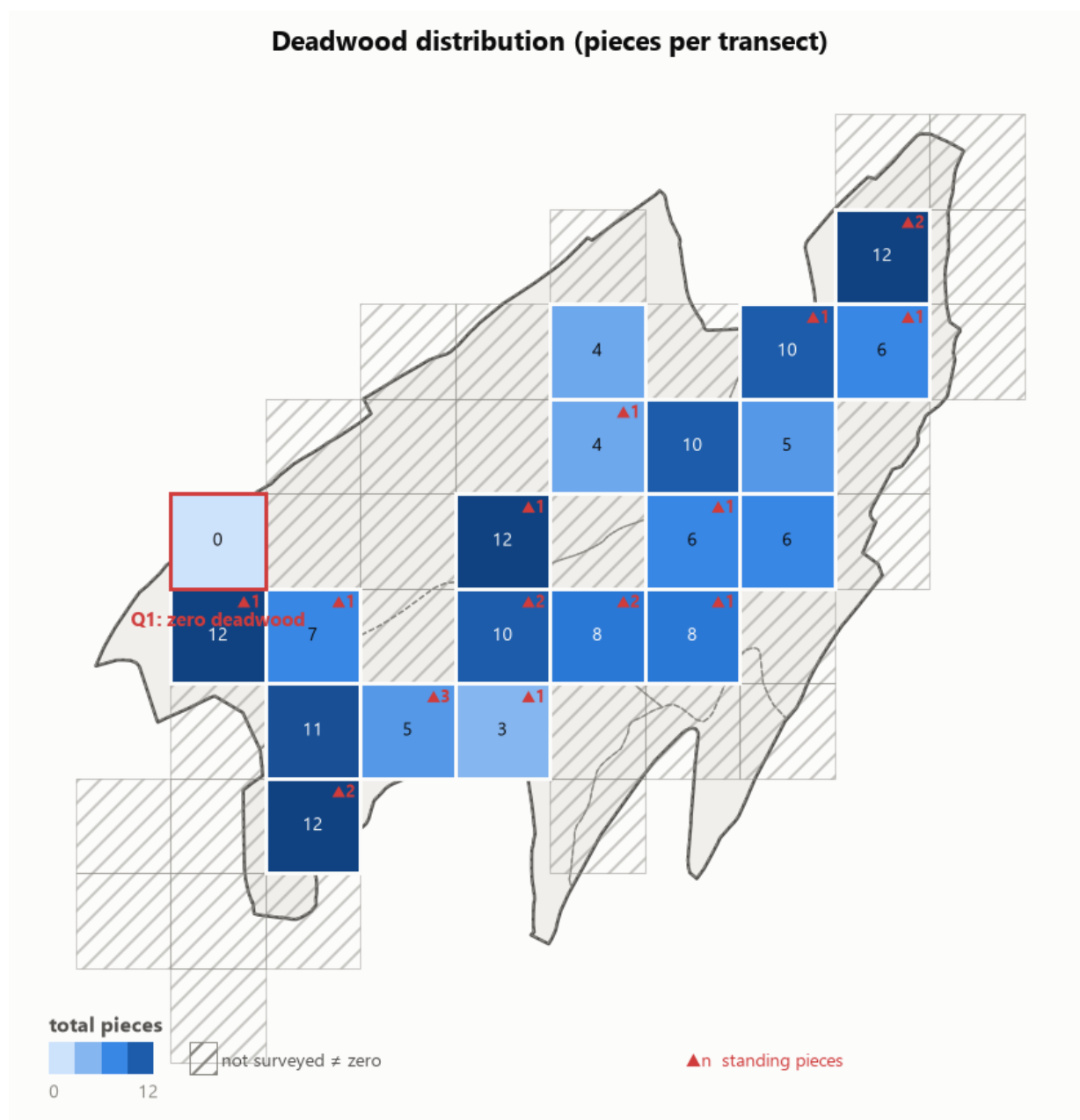


Figure 9 shows how much deadwood each transect held, with red triangles marking the standing component separately from what was lying on the ground.

It is spread fairly evenly across the wood. Fallen wood turned up on 19 of the 20 transects and standing wood on 14, with totals ranging from 0 to 12 pieces. Q1 is the only transect with none of any kind, which might be worth a look on the ground.

Standing deadwood is one of the two things Natural England fail the site on, so this map matters more than most. But the shortage turns out to be about size rather than quantity, **and I'll come back to that in the analysis**

6. Small-leaved Lime and Wild Service Tree

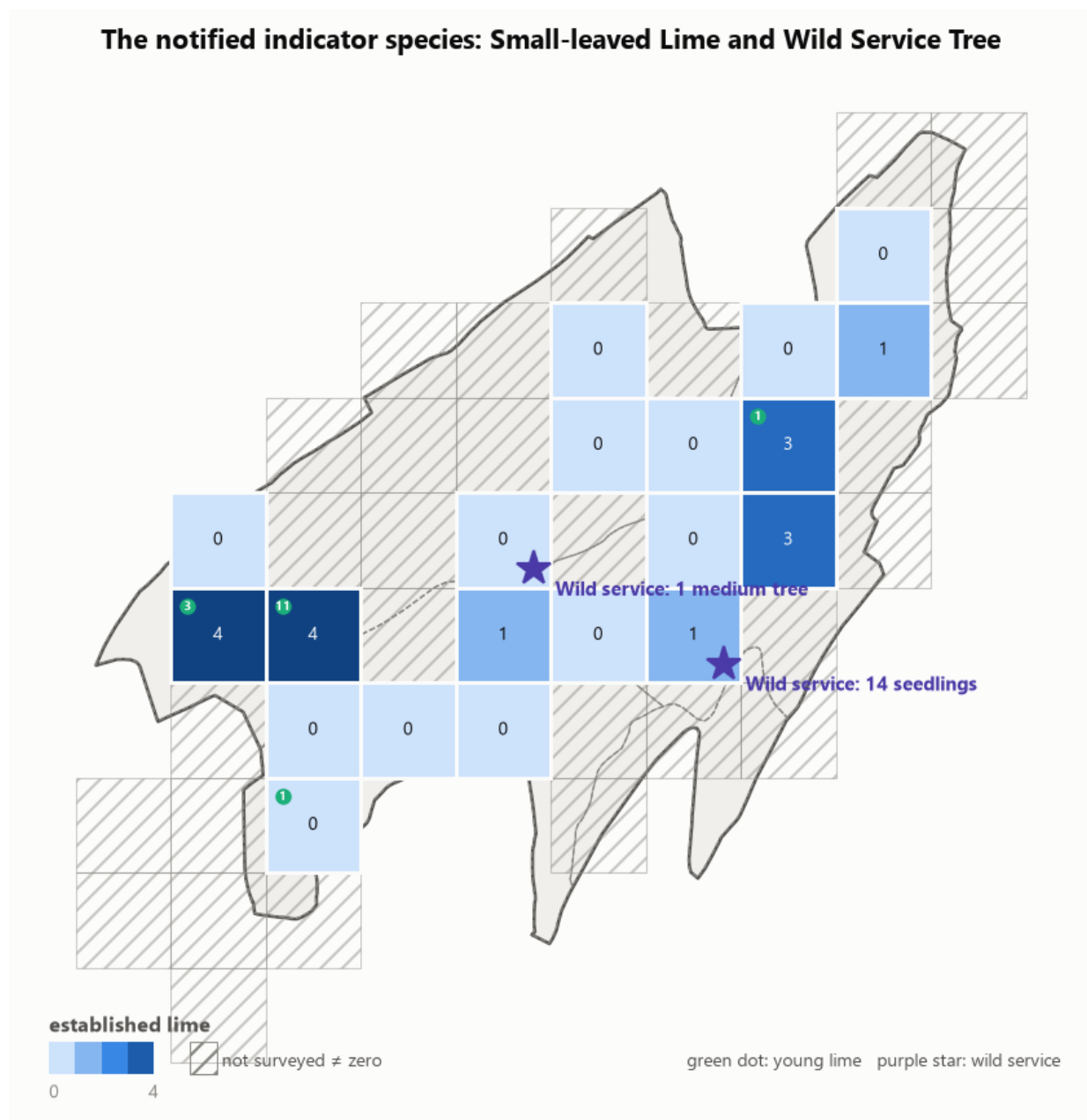


Figure 10 maps the two trees the 1984 citation singles out, both described then as locally abundant.

Small-leaved lime turned up on 8 of the 20 transects, 17 established stems, and 11 of those are in the largest size class. Wild service is thinner: one established tree in the whole survey, plus 14 seedlings on a single transect.

The age structure worries me more than the totals. Eleven overmature stems against six of everything else looks like a population holding on rather than replacing itself.

The young lime are hard to read. They were recorded differently in different places: three on Q2 as coppice, eleven on Q3 as single stems. Since lime rarely sets viable seed here I'd expect most young stems to be vegetative, but I can't tell from the data whether the Q3 records are seedlings or stool regrowth logged another way. It decides whether the lime is recruiting or just persisting. Q3 might be worth a look either way, since eleven young lime is nearly all the young lime on the site, in one 200 m² transect.

I wouldn't read too much into the wild service number on its own. A survey covering 1% of the site could easily miss a clump, so it's evidence of scarcity where we looked rather than proof of loss across the wood. Still seems worth a targeted search.

7. Invasive trees

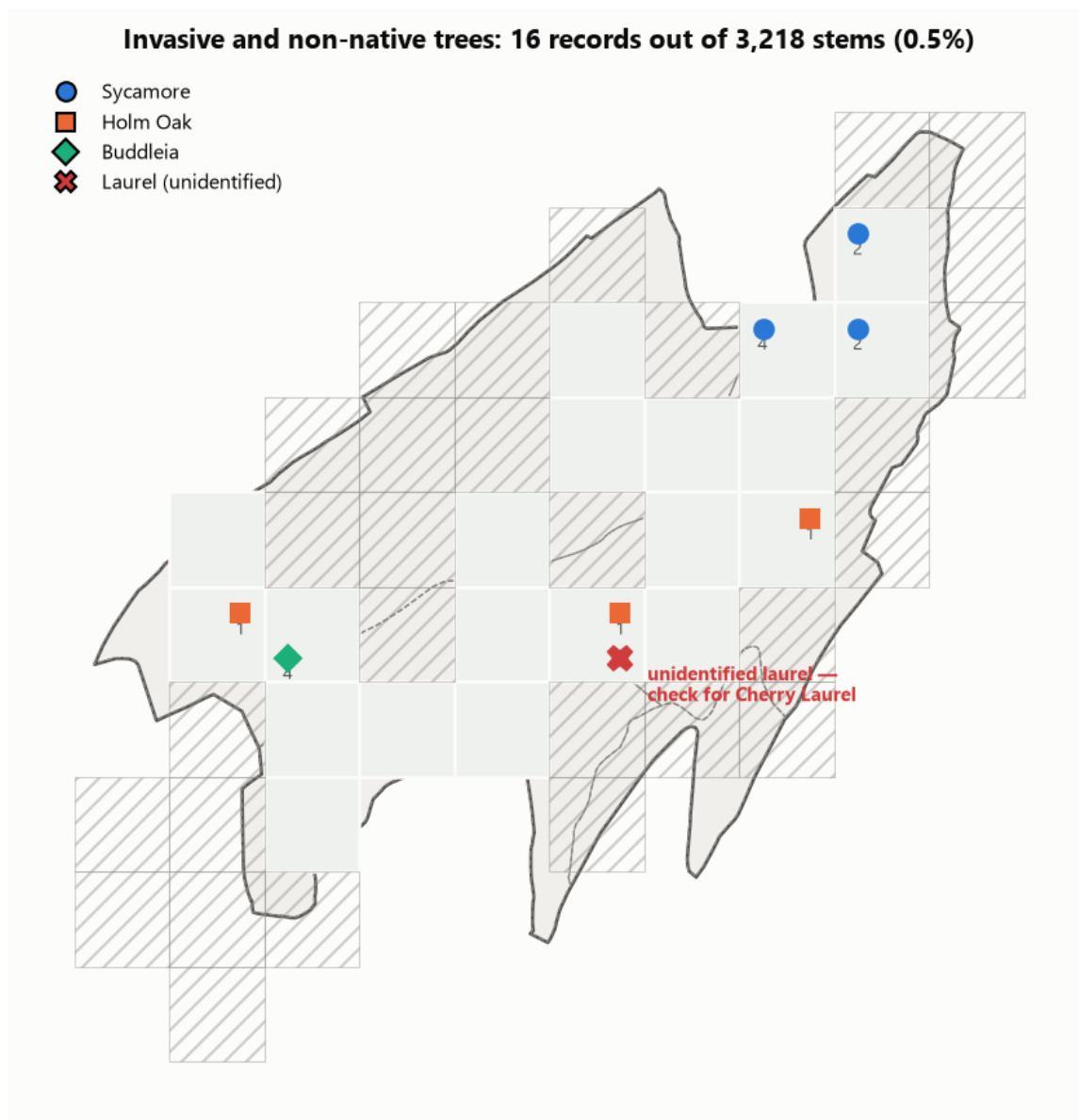


Figure 11 maps the non-native records: 16 stems out of 3,218, so 0.5%. All eight sycamore are in the three most north-easterly transects, next to the housing at North Weston. Q7 and Q15 are on the same eastern side but further south and have none. That might point to seed arriving from gardens rather than an established population in the wood, though eight records is not much to go on.

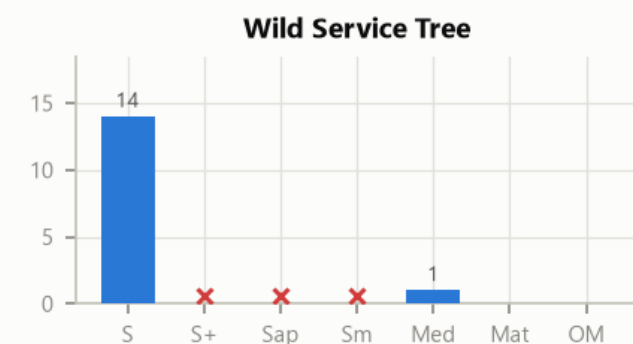
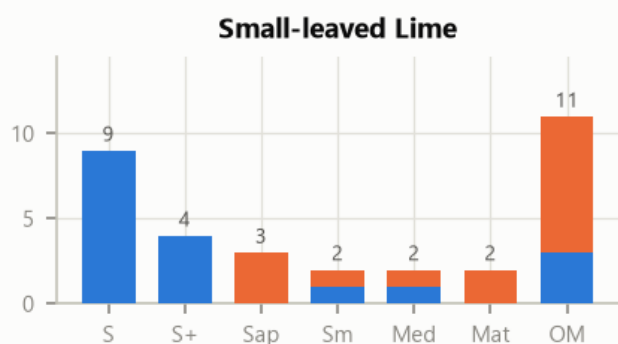
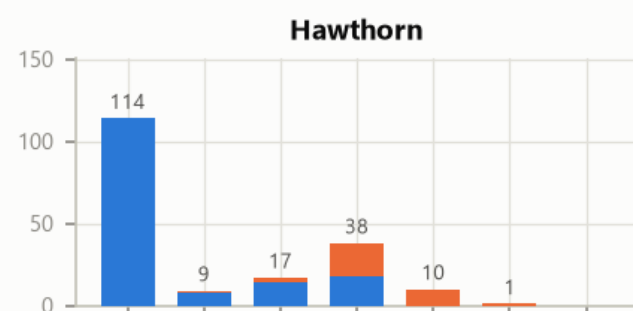
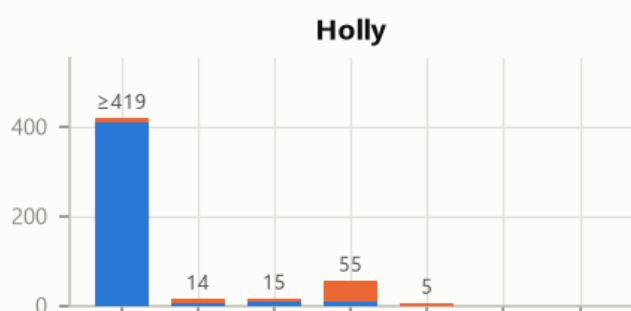
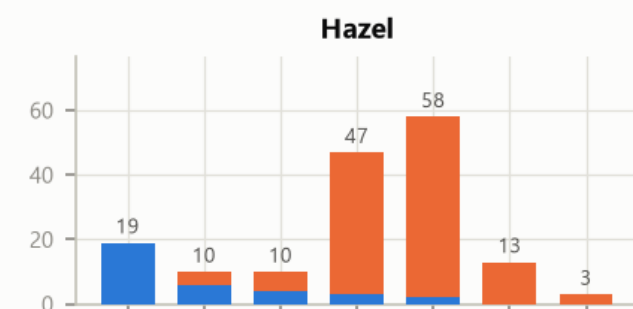
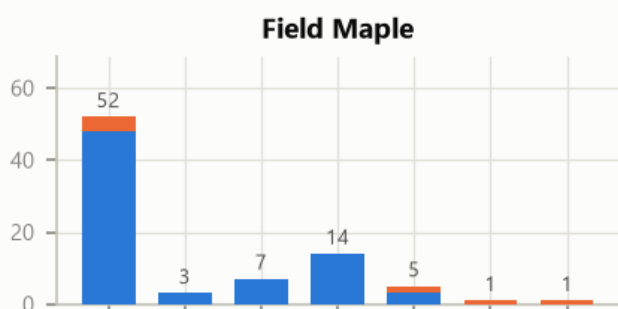
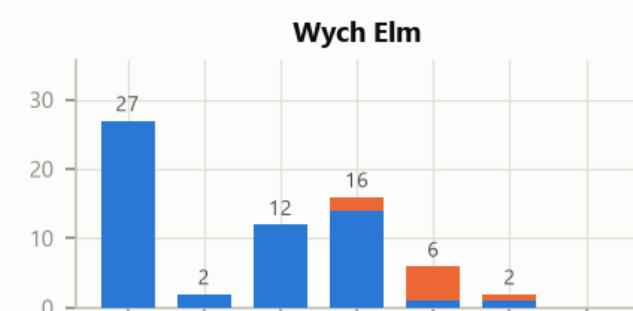
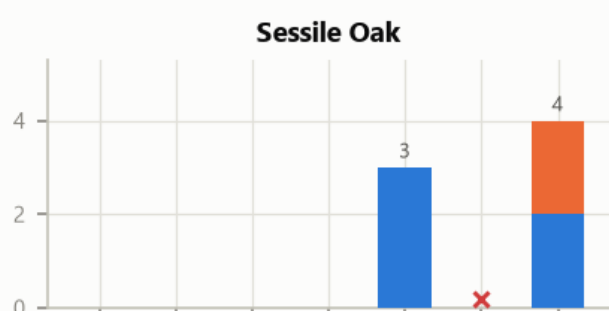
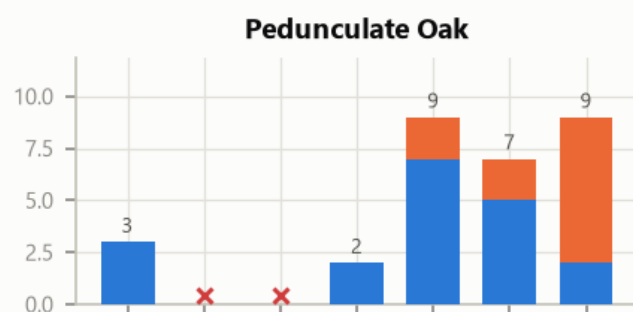
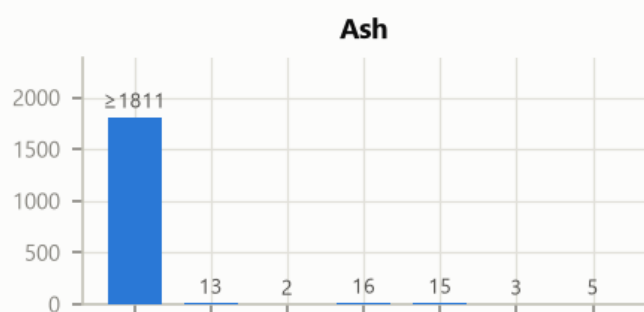
I'm confused by this one. Natural England's 2023 assessment of unit 4 reports sycamore at 15% of the canopy and 80% of the understorey cover at one of their stops. Our survey found eight sycamore stems in the whole wood. Could the difference be because they measure cover and I am counting stems, so a handful of large trees fill the canopy without adding much to a stem count? Or is it more likely our transects simply missed the patch they were standing in?

The single laurel is logged only as "Laurel (*Prunus* sp) Cherry or Portuguese?". If it is cherry laurel that is worth catching early.

Actual analysis

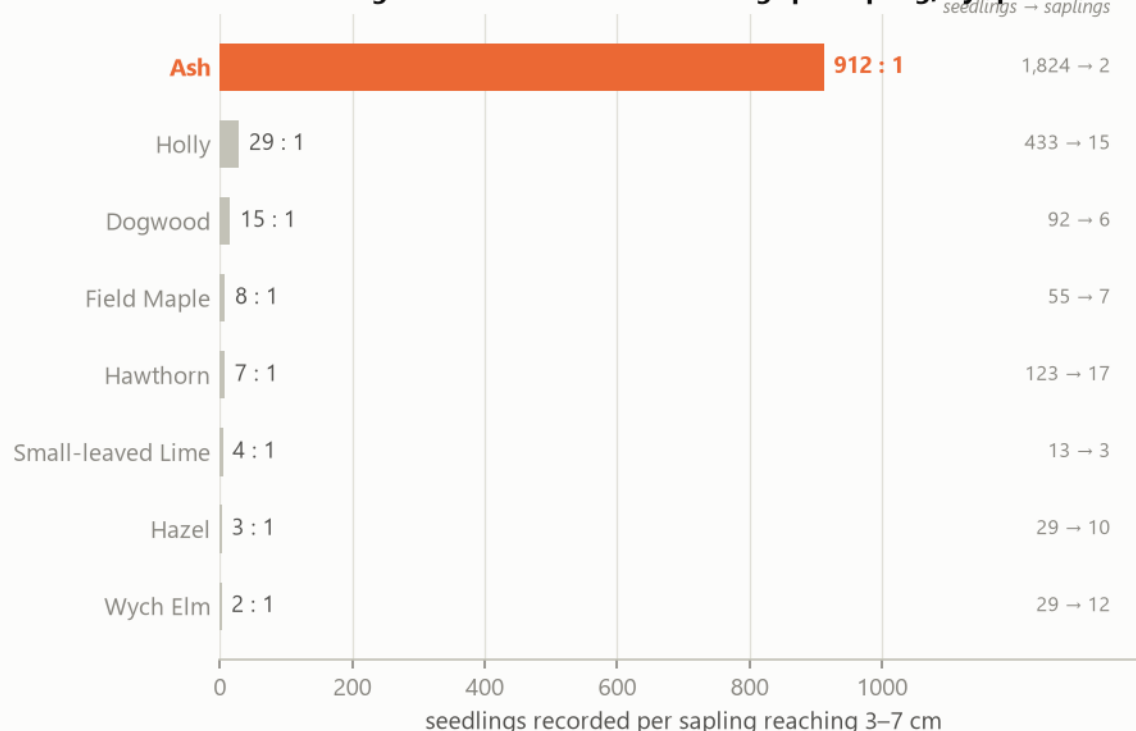
Age structure by species: who is actually in the pipeline

- Single stem (dbh)
- Coppice / multistem (stool width)
- Size class absent, with larger stems present



The regeneration bottleneck: seedlings per sapling, by species

seedlings → saplings



Wild Cherry: 112 seedlings, no saplings · Wild Service Tree: 14 seedlings, no saplings